FIROZ MOHAMMED

DevSecOps Engineer | DevOps Engineer| SRE

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Professional Summary:

- Senior Multi-Cloud DevOps Engineer with 11+ years of experience in Cloud native solutions and Automation.
- Hands on experience in AWS provisioning and good knowledge of AWS services like EC2, VPC, S3, Glacier, ELB, EBS, RDS, Lambda, EKS.
- Experience in SCM, AWS, Environment Management, Continuous Integration (CI), Continuous Delivery (CD) and DevOps Methodologies.
- Configured and managed AWS Certificate Manager (ACM) for certificate issuance, renewal, and deployment.
- Deploying infrastructure on AWS, utilizing services such as EC2, RDS, VPC and Managed Network and Security, Route 53, IAM, CFT, Elastic Beanstalk, S3 and Cloud Watch, Auto scaling, ECS.
- Experience in migrating the Legacy application into AWS platform and managing the AWS services such as Elastic Beanstalk, Lamba,S3,Load Balancing and IAM.
- Proficient in IaC tools like Terraform, CloudFormation, Pulumi, Bicep, and AWS CDK for provisioning and maintenance of infrastructure.
- Built a VPC, established the site-to- site VPN connection between Data Center and AWS.
- Hands-on experience managing Azure applications with Azure App Service Environment, Azure App Service Plan, and Azure Web Jobs, along with expertise in migrating on-prem data to Azure and managing Azure SQL Server databases using CI/CD pipelines for continuous integration.
- Implemented best practices for CI/CD pipeline development, including pipeline security, monitoring, and logging.
- Proficient in Azure automation and security tools (Azure Automation, Azure Policy, Azure Security Center) for compliance and cloud operation management.
- Hands-on experience implementing Palo Alto, including Prisma Access, Prisma Cloud and optionally Prisma Cortex.
- Managed Azure infrastructure, including Azure Web Roles, Azure Worker Roles, Azure SQL Database, Azure Storage, and Azure Active Directory (Azure AD) Licenses. Conducted Azure VM Backup and Recovery using Azure PowerShell and Azure Portal to ensure business continuity.
- Experience in Azure Site Recovery (Azure ASR) for migrating on-premises physical servers and VMware environments into Azure as a disaster recovery solution. Automated the process using Azure PowerShell and Azure Resource Manager (Azure ARM) templates.
- Experience in working on TERRAFORM for provisioning the Infrastructure in AWS, GCP & Azure.
- Experience in migrating the Legacy application into GCP platform and managing the GCP services such as Compute Engine, Big Query, VPC, Stack Driver, Load Balancing and IAM.
- Experience in reducing the cost for compute engines in GCP as per utilization of services.
- Extensive experience with GCP services, including Compute Engine, App Engine, Cloud Storage, Cloud SQL, Cloud Bigtable, and Cloud Pub/Sub.
- Skilled in designing and implementing scalable and secure cloud architectures on GCP
- Experienced with building and managing CI/CD pipelines using GCP services like Cloud Build, Cloud Source Repositories, and Jenkins.
- Hands-on experience in Vulnerability Scanning tools like Nessus, Aqua Security, Trivy, snyk, Prisma Cloud.
- Configured and managed SonarQube instances for multiple projects, ensuring seamless integration with existing CI/CD pipelines.
- Utilized SonarQube to analyze code quality, identifying and addressing technical debt, security vulnerabilities, and coding standards violations.
- Knowledge on Cloud technologies including Infrastructure as a Service, Platform as a Service, and Software as a Service provider (IaaS, PaaS, and SaaS).
- Experience with automation configuration management tools like Ansible.
- Managed and maintained Kubernetes clusters, including node management, scaling, and upgrades.
- Configured and managed Kubernetes services, including load balancing, ingress, and service discovery.
- Worked with Container based deployments using Docker, with Docker images.
- Proficient in monitoring and logging tools like Prometheus, Grafana
- Experienced in using Bug Tracking tools like JIRA and SNOW.

- Good understanding of the principles and best practices of Software Configuration Management (SCM) in Agile, Scrum, and Waterfall methodologies.
- Familiar in Installation, Configuration and Management of RDBMS and NoSQL Databases such as MySQL, PostgreSQL and MongoDB.
- Installed, configured, and managed various Linux distributions, including Ubuntu, CentOS, and Red Hat Enterprise Linux.
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Certifications:

- AWS Certified Solutions Architect Associate (SAA C03)

Education Details:

- Master of Business Administration (MBA) from Jawaharlal Nehru Technological University Hyderabad India (2011).
- Masters in Management Information Systems (MIS) with a concentration in Cybersecurity from Bellevue University Nebraska USA (2024).

Technical Skills:

Cloud Environment	Amazon Web Services (AWS), Azure, GCP (Google Cloud Platform)
Container Security Tools	Trivy, AquaCSPM
Security Standards	CIS Benchmarks, NIST, OWASP
Vulnerability Management	CVE, CVSS
Vulnerability Scanning Tools	Nessus, Aqua Security, Trivy,snyk, Prisma Cloud.
Infrastructure as code	Terraform, Cloud Formation
Application Servers	JBOSS 4.x/5.x/6.x, WebLogic 9.x/10.x, Web Sphere MQ 6.0/7.0
Operating Systems	Linux (Red Hat, CENTOS), Ubuntu, Windows.
Scripting	Java,Groovy,Bash, Shell Scripting and Yaml.
Version Control Tools	GIT
Build Tools	Maven, Gradle and Ant
Containerization Tools	Docker
Orchestration Tools	Kubernetes
Application Servers	WebSphere Application Server, Apache Tomcat, JBoss, WebLogic, Nginx
Integration Tools	EPLX, Jenkins
Automation & Config Tools	Ansible
Networking Protocols	TCP/IP, DNS, DHCP, WAN, LAN, FTP/TFTP, SMTP
Monitoring Tools	AWS Cloud Watch, Dynatrace, Prometheus, Grafana
Bug Tracking Tools	JIRA, SNOW

Work History:

- 4 years -> Secure DevOps Engineer/ Lead, SAP Americas Plano Texas, May 2021-till date.
- 2.5 years -> DevOps Engineer, Tata Consultancy Services, Jul2017-Nov2019.
- 5.4 years -> Site Reliability Engineer, Agile Ideas Pvt Ltd ,Mar 2012-July 2017.

Experience Summary:

Responsibilities as a Secure DevOps Engineer/ Lead:

- Developed and maintained secure Docker images and container configurations, adhering to security policies and guidelines.
- Conducted threat modelling and risk assessments to proactively identify and mitigate security risks in containerized applications.
- Automated security compliance checks using tools like Open Policy Agent (OPA) and integrated them into CI/CD pipelines.
- Installed, configured, and managed SonarQube instances for code quality analysis.
- Utilized SonarQube to analyze code quality, identify bugs, and detect security vulnerabilities.
- Integrated SonarQube with CI/CD pipelines for automated code analysis.
- Designed and implemented CI/CD pipelines using EPLX, ensuring automated build, test, and deployment of applications.

- Utilized GitHub Actions to streamline development workflows, improving efficiency and reducing manual errors.
- Created modular IaC scripts for provisioning VPCs, subnets, VPNs, and security groups, achieving optimized infrastructure management across multiple program areas.
- Utilized GitHub Actions' features, such as matrix builds, caching, and artifacts, to optimize pipeline performance.
- Led the development and implementation of troubleshooting and maintenance procedures, ensuring smooth operations across AWS cloud environments using services such as AWS EC2, AWS Auto Scaling, AWS Elastic Load Balancer, AWS Elastic Beanstalk, AWS S3, AWS CloudFront, AWS RDS, AWS VPC, AWS Route53, AWS CloudWatch, AWS IAM, AWS ECR, and AWS EKS.
- Deployed and managed containerized applications on Kubernetes, including scaling, rolling updates, and self-healing.
- Troubleshooted and resolved Kubernetes-related issues, such as pod failures, network connectivity problems, and resource constraints.
- Created a core platform foundation on AWS using Terraform, setting up Organizations, Accounts, DNS, VPC, Security protocols, VPN, VPC Peering, and VPC-SC Bridge, guaranteeing security and scalability for cloud infrastructure.
- Developed and implemented a cloud network security strategy using security groups, network ACLs, and firewall rules, ensuring secure communication between cloud resources.
- Optimized cloud network performance using techniques such as traffic engineering, load balancing, and content delivery networks (CDNs).
- Provisioned and managed AWS cloud infrastructure using Terraform and Ansible, streamlining configurations and reducing deployment inconsistencies.
- Developed and implemented autoscaling policies and load balancing strategies for AWS Kubernetes clusters (EKS), enabling efficient workload management in dynamic environments.
- Secured AWS Kubernetes clusters through the implementation of RBACs, network policies, and audit logging, ensuring compliance with security best practices and maintaining secure communication with certificates.
- Scaled AWS Kubernetes clusters (EKS) both horizontally and vertically, dynamically adjusting resource usage to optimize performance for applications experiencing varying loads.
- Set up and managed alerting and notification systems using Splunk's alerting framework.
- Collaborated with cross-functional teams to design and implement disaster recovery and data retention strategies for AWS ELK Stack, ensuring data integrity and business continuity for critical applications.

Responsibilities as a DevOps Engineer SRE:

- Deployed applications leveraging the Azure stack, including Azure Virtual Machines (Azure VMs), Azure DNS, Azure Blob Storage, Azure SQL Database, Azure Front Door, Azure Policy, Azure Backup, Azure Cache for Redis,
- Azure Active Directory (Azure AD), with a focus on ensuring high-availability, fault tolerance, and auto-scaling. Enhanced performance and logging using Azure Monitor, Azure Splunk, and Azure New Relic for effective application monitoring.
- Worked extensively with Microsoft Azure services, including Azure Virtual Networks, Azure VPN Gateway, Azure Virtual Machines (Azure VMs), Azure Cloud Services, Azure Resource Groups, Azure Functions, Azure App Service, Azure Load Balancer, Azure Application Gateway, Azure Auto-Scaling, Azure DNS, and Azure Monitor. Integrated Azure Pipelines for seamless Azure CI/CD processes, leveraging Azure GitHub and Azure Repos.
- Provisioned Azure cloud infrastructure using Azure Terraform, and configured environments with Azure Ansible for automation. Expanded deployments using Infrastructure as Code (Azure IaC) with Azure Terraform and Azure Kubernetes Service (Azure AKS) for container orchestration.
- Configured application monitoring systems, setting up Azure Kibana, Azure Logstash, Azure Grafana, and Azure AppDynamics. Integrated Azure Splunk for centralized logging and Azure New Relic for performance monitoring, with special attention to Azure Kubernetes Service (Azure AKS) environments.
- Developed and maintained Azure CI/CD pipelines using tools like Azure Jenkins, Azure Git, Azure Maven, Azure JFrog. Incorporated Azure Pipelines for continuous deployments, utilizing JFrog Artifactory for managing dependencies.
- Wrote Azure Ansible YAML scripts for configuring remote servers and ensured consistent infrastructure through Azure Configuration Management.
- Automated manual tasks and processes to improve governance and compliance using Azure Python. Expanded automation using Azure Bash scripting in both Azure Linux and Azure Windows environments.

- Containerized applications using Azure Docker and deployed them through Azure Kubernetes Service (Azure AKS), managing orchestration and lifecycle with Azure Kubernetes Helm.
- Configured messaging and streaming services using Azure Kafka and Azure RabbitMQ for real-time event processing in Azure microservices architecture.
- Managed web servers, deploying applications on Azure Microsoft IIS and Azure Apache Tomcat, ensuring high scalability and reliability within Azure environments.

Responsibilities as a Site Reliability Engineer:

- Design, implement, and manage scalable, secure, and efficient infrastructure on Google Cloud Platform (GCP).
- Utilized GCP services such as Compute Engine, Kubernetes Engine, and Cloud Storage to design and implement highly availability.
- Design and implement continuous integration/continuous deployment (CI/CD) pipelines using Jenkins.
- Automate monitoring, logging, and alerting using Prometheus & Grafana.
- Implemented monitoring and alerting systems to ensure high uptime and performance, and met or exceeded SLOs.
- Optimized system performance through caching, load balancing, and other techniques.
- Ensure compliance with security and regulatory requirements such as HIPAA, PCI-DSS, or GDPR.
- Implement security best practices for GCP resources, including identity and access management (IAM), network security, and data encryption.
- Conduct regular security audits and vulnerability assessments to identify and remediate potential security risks.
- Monitor cloud-based systems for performance, security, and compliance issues.
- Optimize cloud resource utilization to reduce costs and improve efficiency.
- Mentored to Dev team for setting up infrastructure using PostgreSQL for troubleshooting and Orchestrated using Mesos on Google Cloud Platform (GCP).
- Created, managed, and optimized Docker images for efficient deployment and scaling on GCP.
- Implemented Docker Swarm and Kubernetes to orchestrate containers, ensuring high availability and scalability on GCP.
- Deployed and managed containerized applications on GKE, leveraging its automated scaling, management, and security features.
- Used GCR to store, manage, and secure Docker images, ensuring efficient deployment and scalability on GCP.
- Achieved 99.99% uptime and scalability through Docker-GKE integration and automated scaling on GCP.
- Utilized GitLab as a version control system to manage code repositories, collaborate with team members, and track changes.
- Integrated GitLab with Google Kubernetes Engine (GKE) to automate deployment of containerized applications on GCP.
- Utilized GCP services such as Cloud CDN, Cloud Load Balancing, and Cloud Autoscaling to optimize system performance and reduce latency.